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Studierichting: Informatica
Oefeningenreeks 4A nr. 15.5

$$\int (x^2 - x - 4)e^x dx$$

$$\left\{ \begin{array}{l} u = x^2 - x - 4 \\ dv = e^x dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} du = (2x-1)dx \\ v = e^x \end{array} \right.$$

$$= (x^2 - x - 4)e^x - \int (2x-1)e^x dx$$

$$\left\{ \begin{array}{l} t = 2x - 1 \\ dw = e^x dx \end{array} \right. \Rightarrow \left\{ \begin{array}{l} dt = 2dx \\ w = e^x \end{array} \right.$$

$$= (x^2 - x - 4)e^x - \left((2x-1)e^x - \int 2e^x dx \right)$$

$$= (x^2 - x - 4)e^x - (2x-1)e^x + \int 2e^x dx$$

$$= (x^2 - x - 4)e^x - (2x-1)e^x + 2e^x + c$$

$$= e^x(x^2 - x - 4 - 2x + 1 + 2) + c$$

$$= e^x(x^2 - 3x - 1) + c$$

References